



COVENANT UNIVERSITY

CANAANLAND, KM 10, IDIROKO ROAD
P.M.B 1023, OTA, OGUN STATE, NIGERIA.

B.Sc. DEGREE EXAMINATION

COLLEGE: SCIENCE & TECHNOLOGY

DEPARTMENT: COMPUTER & INFORMATION SCIENCES

SESSION: 2021/2022

COURSE CODE: CSC416

COURSE TITLE: DISCRETE STRUCTURES

SEMESTER: ALPHA

CREDIT UNIT: TWO

INSTRUCTION: ATTEMPT ANY THREE (3) QUESTIONS TIME: 2 HOURS

1. (a) List the members of the following sets.

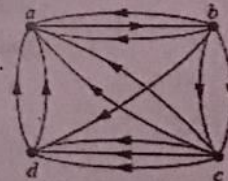
(5marks)

- (i) $\{x \mid x \text{ is a real number such that } x^2 = 1\}$
- (ii) $\{x \mid x \text{ is a positive prime number less than } 20\}$
- (iii) $\{x \mid x \text{ is a perfect square and } x < 100\}$
- (iv) $\{x \mid x \text{ is an integer such that } x^2 = 2\}$
- (v) $\{x \mid x \text{ is an integer such that } 2x = x\}$

- (b) In the Graph diagram below, present in tabular form the following:

(6½marks)

- (i) The number of Vertices
- (ii) The number of edges
- (iii) The total number of In-degree & Out-degree from all the vertices.



- (c) Explain the types of undirected graph, with diagrammatical illustrations.

(12marks)

2. (a) The chairs of an auditorium are to be labeled with an uppercase English letter followed by a positive integer not exceeding 100. What is the largest number of chairs that can be labeled differently?

(6marks)

- (b) Assuming CU has contracted you to help set up simple communication links between the SENATE Building, CUCRID, CLR & CSIS NOC. How many of such simple communication links are possible? Support your answer with a diagrammatical illustration.

(7½marks)

- (c) With the aid of a diagram, explain the following types of function correspondences types:
- (i) One-to-One, not Onto
 - (ii) Onto, not One-to-One
 - (iii) One-to-One and Onto
 - (iv) Neither One-to-One nor Onto
 - (v) Not a Function

(10 marks)

27/8/22

3. (a) Suppose CU sends out convocation invitation cards to graduating students, each student who receives the cards is asked to send them on to four other students. While some students did send out the cards, others did not. If no one receives more than one card and if the card circulation ends after there have been 100 people who read it but did not send it out.

- (i) How many people have seen the card, including the first person? (5marks)
 (ii) How many people sent out the card? (5marks)

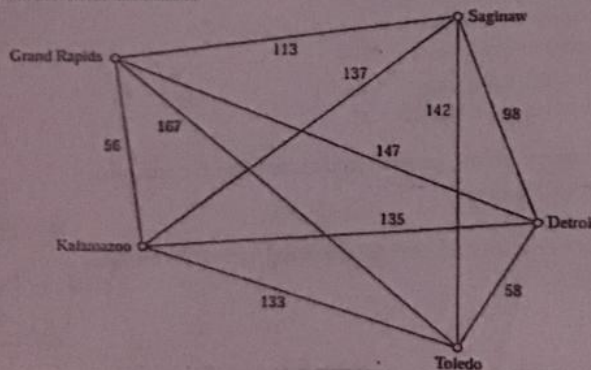
- (b) Highlight, with two relevant examples each, four real-life scenarios where graph models are applicable. (6marks)

- (c) Highlight the various key terminologies of a Rooted Tree. Support your answer with a diagrammatical illustration. (7½marks)

4. (a) Briefly explain the Handshaking Theorem? (5½marks)

- (b) Explain, with examples, what a Balanced M -ary Tree means. (8½marks)

- (c) Given the graph below, a travelling missionary is to embark on a trip from the city of Detroit through all other cities and returns to Detroit, in which order should he visit these cities to travel the minimum distance? (10marks)



5. (a) Highlight, with a relevant example each, four computing domains where graph models are applicable. (6marks)

- (b) In a discrete structures' class, every student is a major in either Computer Science or Maths or both. There of 25 students who major in Computer, while 13 in Maths. If 8 students major in both Computing Science & Maths, how many students are in the entire class? (4marks)

- (c) When is a Vertex said to be *Incident* to an Edge? Show an example (3½marks)

- (d) Construct a truth table for each of the following compound propositions

(i) $(p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$

(5 marks)

(ii) $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$

(5 marks)